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Roll No

ME-5001-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Turbo Machinery

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Differentiate between impulse and reaction turbines. 7
b) Write the application of first law of thermodynamic in turbomachines. 7
2. a) Explain degree of reaction in steam turbine. 6
b) Explain the following terms: 8
 - i) Stator efficiency
 - ii) Carry over efficiency
 - iii) Stage efficiency
 - iv) Vane efficiency
3. In a De-laval turbine, the steam enters the wheel through a nozzle with a velocity of 500m/s and at an angle of 20° to the direction of motion of the blade. The blade speed is 200m/s and the exit angle of the moving blade is 25° . Find the inlet angle of the moving blade, exit velocity of steam and its direction and work done per kg of steam. 14
4. a) What are the advantages of centrifugal pump over reciprocating pump. Explain in brief. 4
b) Classify hydraulic turbines. 4
c) Explain the phenomenon of cavitation in water turbines. 6

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5. A Pelton wheel is having a mean bucket diameter of 1m and is running at 1000rpm. The net head on the Pelton wheel is 700m. If the side clearance angle is 15° and discharge through nozzle is $0.1\text{m}^3/\text{s}$, find: 14
- i) Power available at the nozzle,
 - ii) Hydraulic efficiency of the turbine.
6. a) Draw the vector diagram of centrifugal compressor. 4
- b) Differentiate between the axial flow and centrifugal compressors. 4
- c) Explain working of a centrifugal blower with a neat diagram. 6
7. a) Explain working principle of fluid coupling with the help of a neat diagram. 6
- b) Explain torque ratio, speed ratio, slip and efficiency with reference to turbo machines. 8
8. Write short notes on any four of the following: $4 \times 3\frac{1}{2} = 14$
- a) Torque converter
 - b) Hydraulic intensifier
 - c) Hydraulic press
 - d) Positive displacement pump
 - e) Compounding of turbines
